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Flavour oscillations from non-Hermitian particle mixing

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PHYS40180 Research Project, Literature Review Presentations

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Non-Hermitian Hamiltonians can yield **real** energies.

Bender & Boettcher (1998): Exact symmetry under the combined action of **parity** and **time-reversal** can supersede Hermiticity of the Hamiltonian ($H^\dagger = H$) in quantum mechanics.

Mostafazadeh (2002): This is a special case of **pseudo-Hermiticity**:

$$H^\dagger = \eta H \eta^{-1}, \qquad \eta = \eta^\dagger.$$

Allows to realise phases of **real energy eigenvalues** and **unitary evolution** (conservation of total probability) [Bender, Brody & Jones (2002)] for some non-Hermitian Hamiltonians/non-Hermitian time-independent Schrödinger equations.

Numerous applications, including potential technological impacts across:
[for reviews, see Zyablovsky et al. (2014); El-Ganainy et al. (2018); and Ashida, Gong & Ueda (2021)]

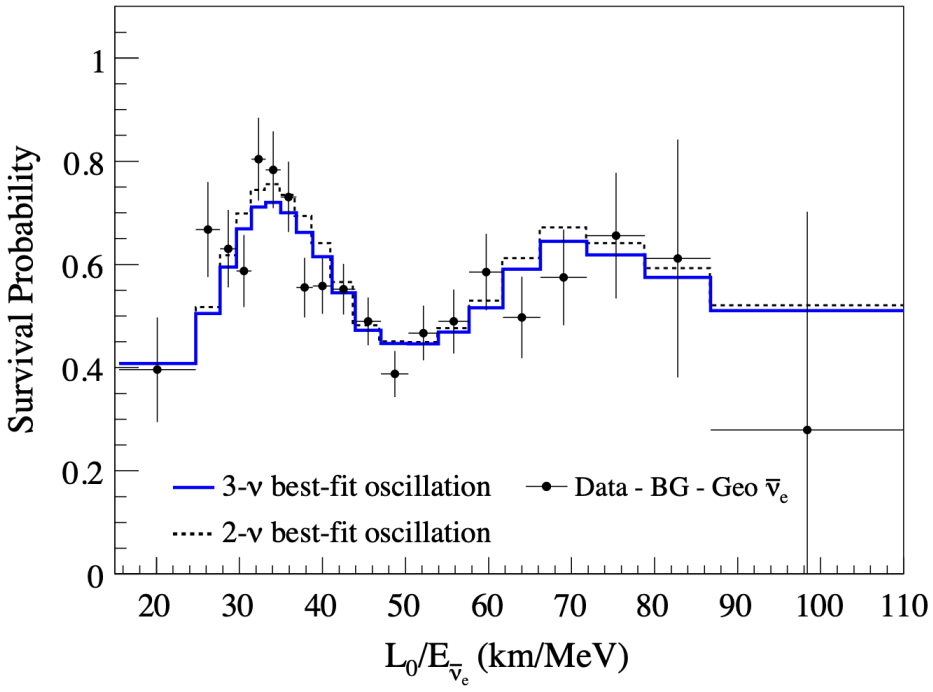
- quantum optics
- photonics
- condensed matter and solid-state physics
- biological physics

Aim: Are **neutrino oscillations** compatible with **non-Hermitian flavour mixing**?

- Neutrinos are produced in **flavour eigenstates** (ν_e, ν_μ, ν_τ), a linear superposition of **mass eigenstates** (ν_1, ν_2, ν_3).
- Relative phases vary during propagation, leading to neutrino oscillations: **neutrinos can change flavour**.
- Described by the **PMNS mixing matrix** [Pontecorvo (1957); Maki, Nakagawa & Sakata (1962)] in Hermitian quantum theory. **Is there a pseudo-Hermitian generalisation?**

Timeliness: Recent progress in constructing **viable** transition probabilities [Alexandre et al. (2026), cf. Ohlsson & Zhou (2020)]

Impact: Pseudo-Hermiticity may offer new ways to build models beyond the Standard Models of particle physics and cosmology, with implications for experiment and observation.



Reproduced from: A. Gando et al. [KamLAND] (2011).

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